

CONFIDENCE IS NOT COMPETENCE: INVERSE-FOLDING MODELS KNOW BINDING IN THE MIXED DERIVATIVE

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ABSTRACT

Staged binder design—generate a backbone, then inverse-fold a sequence—is believed to stumble at protein–protein interface *hotspots*, and a prominent report (ProBID-Net) quantifies this as inverse-folding recovery of 0.334 at hotspots versus 0.472 elsewhere, attributed to dynamics. The phenomenon is real but misread: the model’s binding knowledge was being read from the wrong place. **Confidence and competence are two different derivatives of the same inverse-folding likelihood.** A residue’s *confidence* is the *diagonal* term—a scalar summary of the bound-conditioned distribution—and measures fold-stability constraint; its binding *leverage* is the *mixed second derivative*, the response to ablating the binding partner, $[\log p(a \mid \text{complex}) - \log p(\text{wt} \mid \text{complex})] - [\log p(a \mid \text{monomer}) - \log p(\text{wt} \mid \text{monomer})]$, and measures the binding effect. Because the partner-ablated structure is *determined* by the complex, leverage is computable from the structure but **not** from the bound distribution alone—so every scalar the field reads off the model (recovery, confidence, the complex-vs-monomer KL) is a lossy projection, blind to binding **by construction, not by failure**. This is no vacuous identity: holding the bound distribution fixed, much of the leverage spread survives—a flexible learner (the best of gradient boosting and random forests) trained on the *entire* bound distribution recovers only $\sim 37\%$ of the leverage, so **$\sim 63\%$ is irreducible from the bound distribution** in both inverse-folding families ($\sim 62\%$ even when wt identity is added); the binding-specific component provably requires the partner-ablated second pass. And the consequence is measurable and large. On our main fixture (SKEMPI natural complexes) confidence ranks hotspots at or barely above chance across five architectures and adds nothing beyond geometry (position-level conditional predictive impact 0.000), while the mixed derivative adds binding information **beyond geometry, beyond the standard one-pass log-odds readout, and beyond evolutionary conservation**—the full feature set of published hotspot predictors (mutation-level CPI +0.059; Spearman with experimental $\Delta\Delta G$ -0.30 ; per-position $\sim 5\times$ the best scalar, where confidence is conditionally independent)—and the whole result replicates in three further inverse-folding families (ESM-IF1: CPI +0.035, Spearman -0.26 ; Pi-Fold: +0.050, -0.33 ; MIF: +0.058, -0.27). Three consequences follow. It is **actionable**: the mixed derivative is among the strongest single features for ranking interface hotspots (AUROC 0.69, on par with the learned KL detector at 0.68 and above geometry’s 0.66 and confidence’s 0.51), and adds **+0.016 AUROC [+0.004, +0.029]** on top of the full feature set published predictors already use—geometry and conservation. It is a **dose law** of backbone accuracy—the signal survives ≤ 0.5 Å of error and then collapses, at ~ 1 Å for ProteinMPNN and ~ 1.5 Å for ESM-IF1—and because the sensitivity is to the *backbone the derivative is read from*, not to the network, the fragility (though not its exact threshold, which we measure to be model-dependent) is a prediction every method built on the same mixed derivative (BA-Cycle, RedNet, StaB-ddG) inherits on predicted backbones. And it reaches the **second** mixed derivative: the model’s partner-ablated pairwise couplings predict experimental binding *epistasis*. The published deficit is then largely a corollary—mostly a burial confound that attenuates sharply under matching, across five architectures plus ProBID-Net’s own released model—and

the blindness generalizes from binding to *catalytic* residues (confidence blind, sequence conservation predicts). The leverage operator is BA-Cycle (Jiao et al. 2024); our contribution is the decomposition, the identifiability result and its *measured* non-vacuity, the first beyond-geometry (and beyond-conservation) control on any inverse-folding binding signal, the feature-class law, and—on the intervention side—frozen-model steering by this direction, validated anti-circularly by a second sequence model and by an independent structure predictor. The recipe is general and already familiar: the mixed derivative is the model’s **classifier-free-guidance direction** (the binding partner as conditioner), and to read an untrained quantity off a conditional generative model one ablates the conditioner and reads that direction—not the conditional marginal the field has been mistaking for competence. And the direction is *actionable*: biasing a frozen off-the-shelf ProteinMPNN’s interface logits by $+\alpha \cdot L$ raises the binding-leverage an independent model (ESM-IF1) assigns the sampled residues—monotonically, while a matched-magnitude random direction *lowers* it—with native recovery preserved, and an independent *structure* predictor (AF2-multimer interface ipTM) concurs that the steered interfaces bind better.

1 INTRODUCTION

The dominant paradigm for designing protein binders is staged: a generative model proposes a backbone, and an inverse-folding model assigns a sequence by maximising $p(\text{sequence} \mid \text{backbone})$. This factorisation is convenient but it omits the very quantity a binder exists to optimise—there is no binding-energy term anywhere in $p(\text{sequence} \mid \text{backbone})$. The concern this raises is sharpest at *interface hotspots*: the small set of residues that contribute most of the binding free energy, and which are frequently *frustrated*—buried polar residues or strained rotamers that are locally unfavourable for the monomer’s fold but bought because they pay off in binding. If inverse folding optimises fold-compatibility, the reasoning goes, it should systematically miss exactly these residues, and a hotspot should sit in the tail of the model’s distribution rather than at its mode. A prominent measurement appears to confirm this: ProBID-Net reports inverse-folding recovery of 0.334 at hotspots versus 0.472 elsewhere, and attributes the gap to dynamics.

This paper asks one question: when an inverse-folding model is used to build a binder, where—if anywhere—does its knowledge of *binding* live, and can a designer read it and act on it? The scope is deliberately narrow and practical—structure-conditioned inverse-folding models (ProteinMPNN, ESM-IF1, and three others), protein–protein interface binding, and readouts computable at design time. Our answer is that the binding signal is real, but the field has been reading it from the wrong place: from *scalar* summaries of the model’s output (recovery, confidence, a complex-vs-monomer KL) that are blind to binding **by construction**, when it lives in a **mixed second derivative** that is cheap to compute and directly actionable. The consequence is immediate for practice, in three forms. As a **diagnostic**: inverse-folding confidence is a geometry detector, not a binding score, so a pipeline that distrusts it at the interface (as the field’s leading one-shot designer does, by freezing the interface) is right to—and can now do better. As a **training-free ranker**: the mixed derivative locates interface hotspots and adds on top of the exact feature set published predictors already use, at no training cost. As a **drop-in steering knob**: a $+\alpha \cdot L$ tilt biases a *frozen, off-the-shelf* inverse-folding model toward higher-binding interface residues with no retraining, confirmed by both an independent sequence model and an independent structure predictor. And the recipe is not protein-specific—the mixed derivative is the model’s **classifier-free-guidance direction**, so the same move (ablate the conditioner, read the mixed derivative, not the marginal) applies to any conditional generative model, which is what places the result at a representation-learning venue rather than a purely methodological one.

There is a confound that no prior analysis controls, and it runs opposite to intuition. Hotspots are, on average, *more deeply buried* than other interface residues, and burial is precisely where inverse folding is *most* confident and accurate. An uncontrolled hotspot-vs-rest comparison therefore mixes a putative binding effect with a large burial effect of the opposite sign—so the naive comparison **hides** any real deficit rather than inventing one. Controlling burial is not a detail; it is the experiment.

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